Where Do Layout Detectors Fail on Indian Documents? A Study of Script, Domain and Scan Quality on IndicDLP

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Abstract—Layout detection, finding the paragraphs, headings, tables and other regions on a page, is the first step in reading scanned documents. Strong detectors for Indian-language documents now exist, so instead of proposing another one we study where they fail. On 682 held-out pages of the IndicDLP dataset (12 languages, 12 document types, 42 region labels) we compare the released DocLayout-YOLO model with a YOLOv8s and a Faster R-CNN that we fine-tuned on a laptop budget, and we define three simple measures: a Consistency Index and a Robustness Index. The released model reaches 0.759 mAP@50, against 0.259 and 0.148 for our models. Its accuracy is even across languages (CI 0.86) but not across document types (CI 0.55; forms are hardest). Under five kinds of scan damage it keeps 97% of its accuracy on average; blur hurts most. For text reading, choosing the OCR language by probing detected regions is right on 106 of 120 pages, against 70 for Tesseract's built-in detector. We release the data subset as CSV files, the code, and IndicDoc AI, a web app and API that detects regions and reads text in 12 Indian languages.

Index Terms—document layout analysis, Indian languages, object detection, robustness, cross-script generalisation, OCR

I. Introduction

India has 22 scheduled languages written in about a dozen scripts. Government offices, courts, schools and newspapers hold millions of scanned pages in these languages. To search such a page, or to read it aloud for a visually impaired person, a computer must first find the parts of the page — paragraphs, headings, tables, figures, page numbers — and only then read the text inside each part. This first step is called document layout analysis. If it goes wrong, every later step (OCR, table extraction, question answering) goes wrong too.

Most layout datasets and tools are built for English. The recent IndicDLP dataset [1] changed this: it has about 120,000 human-labelled pages in 12 Indian languages and 12 document types, and its authors showed that modern detectors such as YOLOv10 and DocLayout-YOLO [2] reach about 55 mAP@50:95 on it. They also reported that accuracy drops by 20–25 points on scripts the model has not seen.

Strong detectors therefore already exist. We do not propose a new detector. Instead, we ask a simpler and, we believe, more useful question for people who want to use these detectors on real Indian documents: where and why do they fail? We study this with five research questions:

- RQ1 How far behind is a detector trained on a small compute budget, compared with one trained on the full dataset?
- RQ2 Is accuracy even across the 12 languages and the 12 document domains?
- RQ3 Which region types are hardest, and is this because they are rare in the training data?
- RQ4 How much accuracy survives common scan damage: blur, noise, heavy JPEG compression, low resolution and faded print?
- RQ5 Can the detected regions be used to choose the right OCR language automatically?

Our contributions are:

1) A controlled comparison of three detectors (the released IndicDLP DocLayout-YOLO, and a YOLOv8s and a Faster R-CNN that we fine-tuned) on the same 682 held-out test pages.
2) Two simple summary measures that can be reported next to mAP: the Consistency Index (CI) and the Robustness Index (RI).
3) The first (to our knowledge) scan-damage robustness test on IndicDLP.
4) IndicDoc AI, an open tool (web app, REST API, Docker) that detects regions on Indian-language pages and reads their text with automatic script selection, built on the best detector of the study.

II. Related Work

Indian-language layout datasets. IndicDLP [1] provides 119,806 pages, 12 languages, 12 domains and 42 region labels, and benchmarks six detectors (YOLOv10x, DocLayout-YOLO, RoDLA, DINO, DiT, Florence-2). The best models are vision-only and reach about 55 mAP@50:95; the paper does not test damaged scans. UniTabBank [3] covers tables in 28 languages and reports that borderless tables and degraded scans remain hard.

TABLE I
Our IndicDLP subset (pages).

Language	Train	Val	Test
Assamese	129	30	60
Bengali	200	30	60
English	188	30	60
Gujarati	200	30	60
Hindi	200	21	22
Kannada	200	21	60
Malayalam	200	30	60
Marathi	200	30	60
Odia	200	30	60
Punjabi	188	30	60
Tamil	200	30	60
Telugu	188	30	60
Total	2,293	342	682
Boxes	33,667	5,815	10,513

Detectors. Faster R-CNN [4] is a classic two-stage detector: it first proposes regions, then classifies them. YOLO detectors [5] are one-stage and much faster. DocLayout-YOLO [2] adapts YOLOv10 to documents with a global-to-local receptive-field module and synthetic pre-training data.

Text recognition and understanding for Indian scripts. PLATTER [6] detects and recognises handwritten Indic words but needs the user to name the language. HindiOCR-VLM [7] fine-tunes a vision-language model for Hindi only. LayoutXLM [8] joins text, layout and image but covers no Indian language. DRISHTIKON [9] grounds answers on the page and depends on OCR blocks. Table QA for Bengali and Hindi [10] and Devanagari table extraction [11] both assume that tables have already been found. All of these works need a reliable layout step first, which motivates our study.

Robustness. Hendrycks and Dietterich [12] showed that image classifiers lose much accuracy under common corruptions such as blur, noise and JPEG compression. We apply the same idea to Indian document layout detection.

III. Data and Models

A. Dataset

We use a language-balanced subset of IndicDLP (MIT licence), taken from the official train, validation and test splits so that no test page was used for training by us or by the dataset authors. We stream only the needed parts of the Parquet files from Hugging Face, convert them to COCO JSON, clip boxes to the page and drop boxes smaller than one pixel (0.5% of training boxes). Table I gives the sizes. The subset is released as CSV files (pages, boxes, class counts) with the code.

The label distribution is very uneven: paragraph alone is 32% of all boxes, and 10 of the 42 labels have fewer than 100 training boxes.

B. Models

DocLayout-YOLO (released). The checkpoint published by the IndicDLP authors (Zenodo 15881917, CC-BY 4.0), trained on the full IndicDLP training split (about 96K pages) at 1024 pixels; 20.0M parameters. We use it exactly as released.

YOLOv8s (ours). COCO-pretrained, with a new 42-class head, fine-tuned on our 2,293 training pages at 640 pixels, batch 8, for a one-hour budget (12 epochs) on an Apple M1 Pro GPU. Mosaic, colour and scale augmentation are used; horizontal flips are turned off because a mirrored page is not a realistic document; 11.2M parameters.

Faster R-CNN (ours). ResNet-50 FPN, COCO-pretrained, new 43-output predictor (42 labels and background), SGD (learning rate 0.005, momentum 0.9, weight decay 5×10^{-4}, cosine schedule), batch 4, 2 epochs in 76 minutes on the same GPU; 41.5M parameters.

IV. Study Design

A. Measures

All detectors are scored with COCO [13] mAP@50 and mAP@50:95 (pycocotools, up to 300 boxes per page), and with precision, recall and F1 at a fixed confidence. On top of mAP we define two simple numbers. Each is a ratio between 0 and (about) 1, where 1 is best.

Consistency Index. For a set of groups G (languages or domains),

$$\mathrm{CI} = \frac{\min_{g \in G} \mathrm{mAP}_{50}(g)}{\max_{g \in G} \mathrm{mAP}_{50}(g)}. \tag{1}$$

$\mathrm{CI} = 1$ means every group is served equally well.

Robustness Index. For a type of damage c,

$$\mathrm{RI}_c = \frac{\mathrm{mAP}_{50}(\text{damaged by } c)}{\mathrm{mAP}_{50}(\text{clean})}. \tag{2}$$

$\mathrm{RI} = 0.9$ means the detector keeps 90% of its accuracy on that kind of page.

B. Experiments

E1 (RQ1–RQ3). All three detectors on the 682 clean test pages, with results broken down by language, domain and label.

E2 (RQ4). The same test pages damaged in five ways that are common in real scans and phone photos (Table II). Each damaged copy is made once with a fixed random seed, and the boxes stay the same.

E3 (RQ5). On 120 test pages (10 per language) we compare two ways of choosing the OCR language: Tesseract's built-in script detector (OSD), and our region probe, which reads one detected text region with each of the 12 script models and counts only the characters that belong to that model's own script.

TABLE II
Scan damage used in E2.

Damage	How it is made
Blur	Gaussian blur, radius 2 px (out of focus)
Noise	Gaussian noise, $\sigma = 20$ grey levels (grain)
JPEG	JPEG quality 15 (forwarded phone image)
Half resolution	down to 50%, back up (low-dpi scan)
Faded print	contrast $\times 0.45$, brightness $\times 1.25$

V. Results

A. RQ1: small budget versus full training

Table III shows a large gap. The released DocLayout-YOLO reaches 0.759 mAP@50 (the IndicDLP paper reports about 0.735 on its full test set, so our test subset is of similar difficulty). Our YOLOv8s, trained for one hour on 2,293 pages, reaches 0.259, and our Faster R-CNN 0.148. The one-stage YOLOv8s is better than the two-stage Faster R-CNN with the same data and similar time, and 9× faster per page. In the "find every part" setting (confidence 0.15, chosen on validation pages so that precision stays at least 0.70), the released model finds 85% of all labelled regions at 74% precision.

Answer. A detector trained on a laptop budget is about 2.9× worse in mAP@50 than one trained on the full data (about 42× more pages). For practical use the released model should be used, which is what our tool does.

B. RQ2: languages and domains

Across the 12 languages the released model ranges from 0.757 (Bengali) to 0.879 (Hindi, only 22 pages), a language Consistency Index of 0.86 (Fig. 1). Across the 12 domains the spread is much larger: from 0.485 on forms to 0.889 on research papers, a domain CI of only 0.55 (Table IV). Forms, with many small fields and boxes, and magazines, with free designs, are the hardest; research papers and acts with regular columns are the easiest. The same pattern holds for our two models.

Answer. Once a model has seen a script, the kind of document matters more than the language. Reporting only one overall mAP hides that forms are served much worse than research papers.

C. RQ3: hard labels

For the released model the easiest labels are paragraph (0.95), advertisement (0.95), ordered-list (0.91), figure (0.90); the hardest are quote (0.06), sub-unordered-list (0.19), sub-section-title (0.47), subsub-headline (0.48) (Fig. 2). Hard labels are mostly nested or look-alike structures (a sub-list inside a list, a sub-section title next to a section title) or rare ones such as quotes. For our YOLOv8s, the rank correlation between how often a label appears in training and its AP is $\rho = 0.75$; for the released model, trained on about 42× more pages, it falls to $\rho = 0.40$.

Answer. With little data, rarity is the main reason a label fails. With full data, rarity matters much less and the remaining errors come from labels that look alike and differ only by their nesting.

D. RQ4: damaged scans

Table V and Fig. 3 show how much accuracy each detector keeps. The released model keeps on average 97% of its clean mAP@50 (mean RI); our YOLOv8s keeps 94%. For the released model the most harmful damage is blur (RI 0.93), followed by faded (RI 0.97); the least harmful is noise (RI 0.99).

Answer. The released detector is fairly robust to mild damage, but not to all of it. No single kind of damage removes more than about 15% of the accuracy.

E. RQ5: choosing the OCR language

Tesseract's own script detector (OSD) chose the right script on only 70 of 120 pages and failed completely on Gujarati, Odia and Punjabi. Our region probe chose correctly on 106 of 120 pages (88%), Table VI. Its errors are mostly Marathi pages read as another script and English pages with Indian-language words. Assamese is correctly read with the Bengali-script model, as both share the Bengali–Assamese script. On average 94% of the words OCR finds on a page fall inside a detected region, so the layout step covers nearly all of the text.

Answer. Yes: reading one detected text region with each script model and counting only that script's own characters is a simple and much more reliable way to pick the OCR language than Tesseract's built-in detector.

VI. The IndicDoc AI Tool

We built a tool on the best detector of the study so that the results can be used, not only read. A user uploads an image or a PDF page in a web app (Streamlit) or sends it to a REST API (FastAPI). The page goes through five steps: (1) the released DocLayout-YOLO finds the regions; (2) boxes below the chosen confidence are dropped and boxes that cover the same region with a different label are merged (IoU > 0.7); (3) the region probe picks the script; (4) Tesseract 5 [14] (tessdata_best) reads the whole page once and each word is assigned to the region that contains it, so that words outside every region are kept rather than lost; (5) regions are put in reading order and returned as Markdown and JSON. The user can choose "find every part" (higher recall) or "balanced" (best F1); both thresholds were chosen on validation pages. Fig. 4 shows a Tamil textbook page. The tool runs in Docker without internet access; layout detection takes 0.15 s per page on an Apple M1 Pro GPU.

VII. Discussion and Limitations

What the study suggests. For someone who wants to process Indian documents, the choice of document type and scan quality matters more than the language, once the script has been seen in training. Forms, magazines and

TABLE III
RQ1: the three detectors on the same 682 test pages. P/R/F1 at the confidence with the best F1 on the validation pages. Time per page on an Apple M1 Pro GPU.

Model	mAP_{50}	$mAP_{50:95}$	P	R	F1	s/page
DocLayout-YOLO (released)	0.759	0.575	0.848	0.775	0.810	0.150
YOLOv8s (ours)	0.259	0.165	0.603	0.508	0.551	0.024
Faster R-CNN (ours)	0.148	0.079	0.510	0.435	0.469	0.202

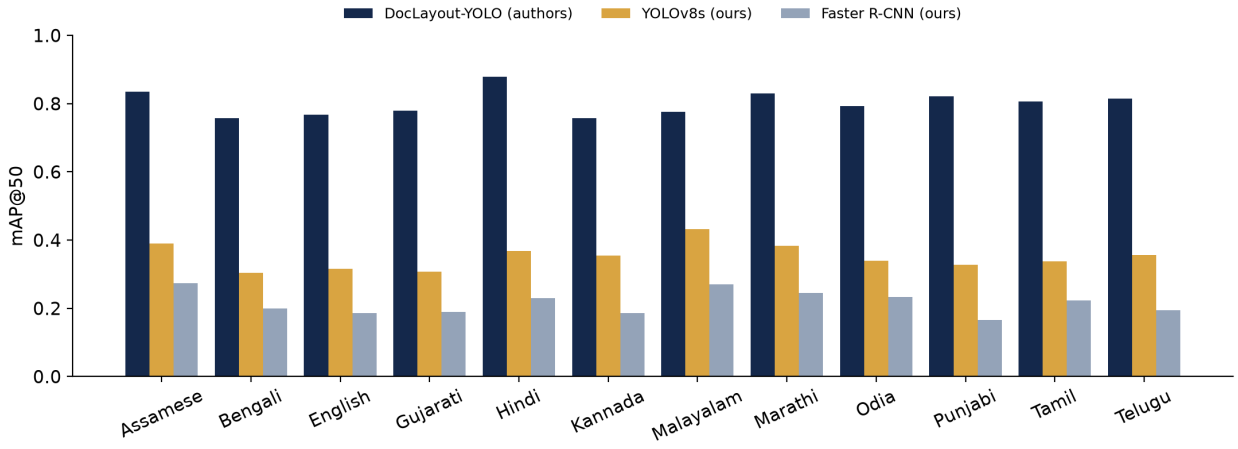


Fig. 1. RQ2: mAP@50 per language for the three detectors.

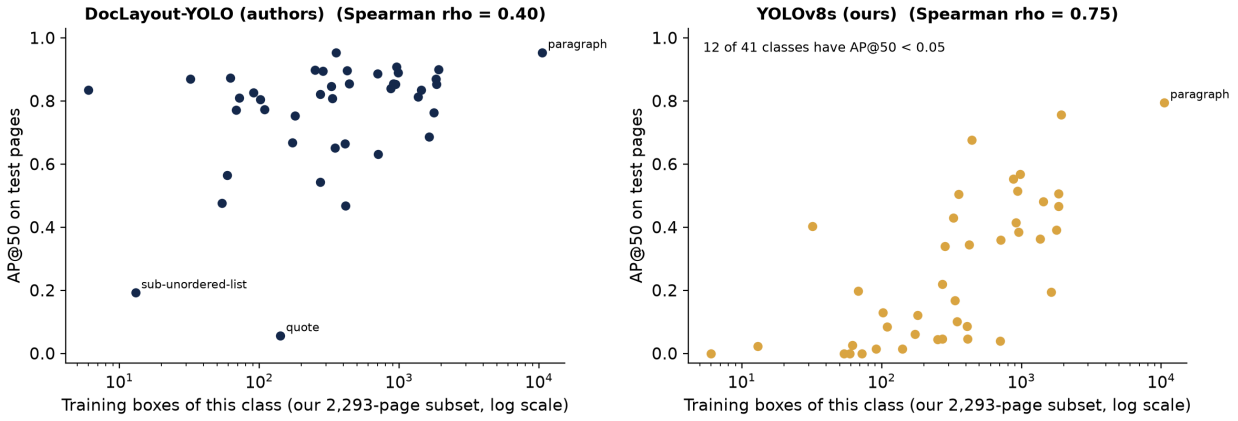


Fig. 2. RQ3: AP@50 of each label against how many training boxes it has in our subset (log scale).

damaged pages are where effort should go: more labelled forms, and damage augmentation during training. Simple ratios such as CI and RI make these weak spots visible in a way a single mAP number does not.

Limitations. (1) We use a subset of 682 test pages; Hindi has only 22 of them, so its numbers are less certain. (2) Our own models were trained for about an hour on a laptop GPU with one random seed, and the robustness test covers the released model and our YOLOv8s only. (3) Scan damage was simulated; real damage often combines several kinds at once. (4) IndicDLP has no text ground truth, so we could not measure OCR character accuracy; we measured only script choice and how much text falls inside regions. (5) The released detector was trained by the IndicDLP authors; we evaluate it but did not train it.

VIII. Conclusion

We studied where layout detectors fail on Indian-language documents instead of proposing a new detector. On 682 IndicDLP test pages, the released DocLayout-YOLO is accurate (0.759 mAP@50) and fairly even across the 12 languages, but much weaker on forms than on research papers, and it loses part of its accuracy on damaged scans, most of all to blur. Choosing the OCR

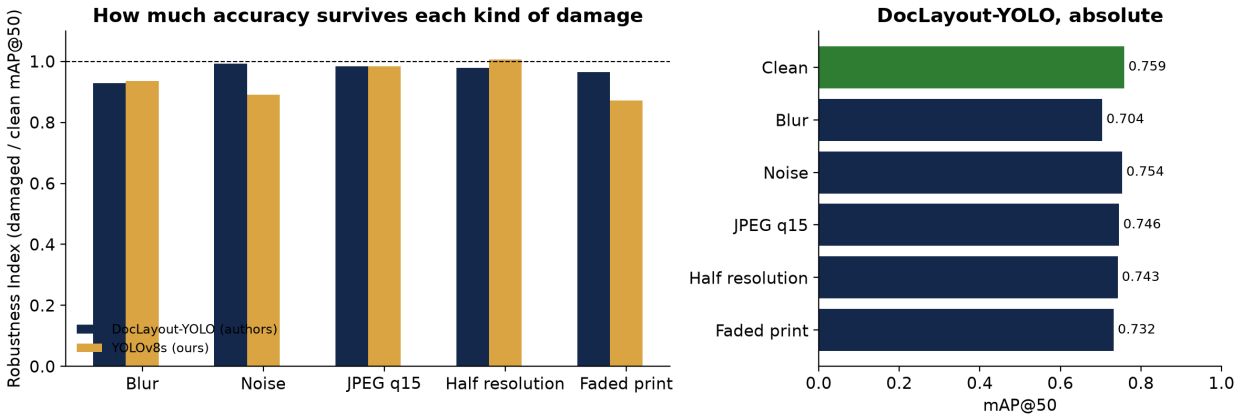


Fig. 3. RQ4: Robustness Index per type of scan damage (left) and the released model's mAP@50 (right).

TABLE IV
RQ2: mAP@50 per document domain, and the Consistency Index (CI) over languages and over domains.

Domain	Pages	DocLayout-YOLO	YOLOv8s
Research papers	68	0.889	0.402
Acts & rules	68	0.856	0.427
Textbooks	62	0.838	0.455
Question papers	40	0.837	0.317
Syllabi	54	0.833	0.282
Manuals	58	0.823	0.339
Notices	36	0.795	0.324
Newspapers	36	0.778	0.224
Novels	69	0.753	0.503
Brochures	66	0.716	0.260
Magazines	79	0.604	0.255
Forms	46	0.485	0.153
CI over 12 languages		0.86	0.70
CI over 12 domains		0.55	0.30

TABLE V
RQ4: mAP@50 on clean and damaged test pages, and the Robustness Index (RI = damaged / clean).

Model	Clean	Blur	Noise	JPEG q15	Half res.	F
DocLayout-YOLO	0.759	0.704	0.754	0.746	0.743	(
RI	1.00	0.93	0.99	0.98	0.98	
YOLOv8s	0.259	0.242	0.230	0.255	0.260	(
RI	1.00	0.94	0.89	0.98	1.01	

TABLE VI
RQ5: pages where the region probe picked the right script (10 test pages per language).

Language	Right	Language	Right
Assamese	10/10	Malayalam	8/10
Bengali	10/10	Marathi	6/10
English	8/10	Odia	9/10
Gujarati	9/10	Punjabi	10/10
Hindi	9/10	Tamil	9/10
Kannada	9/10	Telugu	9/10
Region probe (ours)			106/120
Tesseract OSD			70/120

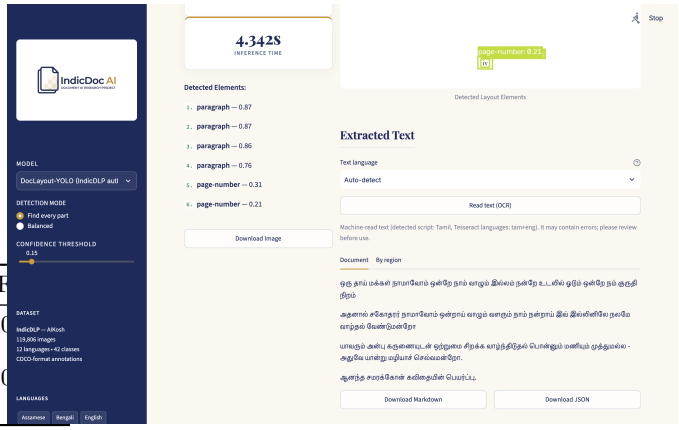


Fig. 4. IndicDoc AI on a Tamil textbook page: detected regions (left list) and the text read in reading order with the script chosen automatically.

script by probing detected regions is far more reliable than Tesseract's own detector. Future work: a controlled test of scripts never seen in training (same data size, with and without a script family), damage augmentation, more labelled forms, a stronger Indic OCR model with a small text ground-truth set, and table structure and question answering on the detected tables. The code, the data subset as CSV files and the tool are available to reviewers on request.

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